

Fig. 5: Actual-size, single-side PCB for alphanumeric display

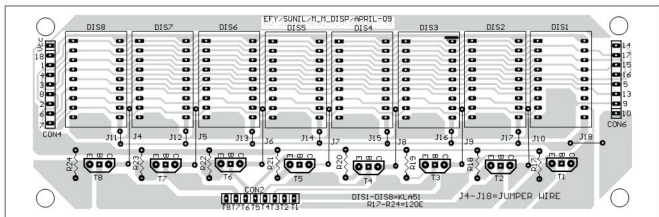


Fig. 6: Component layout for the PCB shown in Fig. 5

ponent layout for this PCB is shown in Fig. 4.

Fig. 5 shows the PCB for displays DIS1 through DIS8, transistors T1 through T8 and resistors R17 through R24. Component layout for this PCB is shown in Fig. 6. You need to use an additional PCB as shown in Fig. 5 for DIS9 through DIS16, so as to configure 16 alphanumeric displays. For this PCB, the corresponding components will be transistors T9 through T16 and resistors R25 through R32 in addition to displays DIS9 through DIS16. Corresponding connector are provided to make a proper connection. Connectors CON2, CON4 and CON6 of Fig. 3 are connected to CON2, CON4 and CON6 of Fig. 5, respectively, through external wires to interface DIS1 through DIS9. Connectors CON3, CON5 and CON7 of Fig. 3 are connected to CON2, CON4 and CON6 of Fig. 5, respectively,

through external cable to interface DIS9 through DIS16.

Software and its working

The source code 'movmsg.asm' is written in Assembly language and assembled using cross-compiler. It is well-commented and easy to understand. Timer 1 has been used to generate a delay of around 1 ms for the switching gap between two consecutive displays. Thus, each display is enabled for 1 ms while displaying a message. The length of this cycle depends upon the length of the message string. The cycle repeats after a '0' is encountered at the end of each message stored in the look-up table at the end of the program.

Each time, to display a character at a given display, first two bytes (16 bits) of data are sent to Port-2 and Port-0, then the desired display is enabled by sending its address to Port-1. Thereaf-

ter, a delay of 1 ms (slightly more than that) is generated by timer 1. Upon timer overflow, the entire display panel is refreshed by passing 'FFFFH' to the data bus. Then the next character at the next display is passed in the similar manner. The cycle frequency is variable (depending upon the length of the message) but always high enough so that the message appears continuous to the human eye.

Timer 0, with its interrupt enabled, is used to change the starting address of the message in cyclic manner so that the characters scroll from left to right with a proper gap between each shift. Meanwhile, the interrupt service sub-routine also checks for the starting address of DIS16 (right-most display). As soon as the first character reaches DIS16, the message stays for a longer time so that the entire message (message length not longer than 16